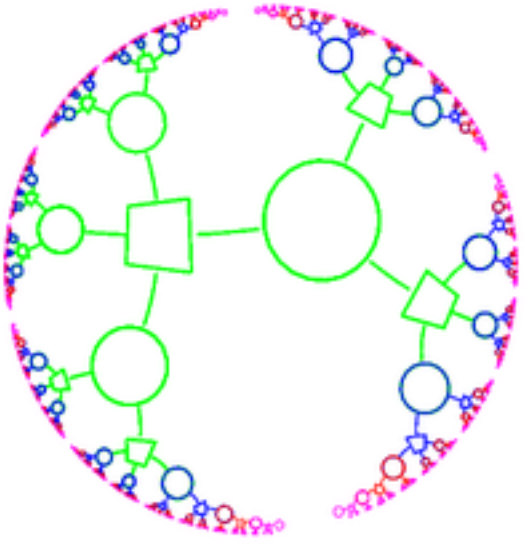
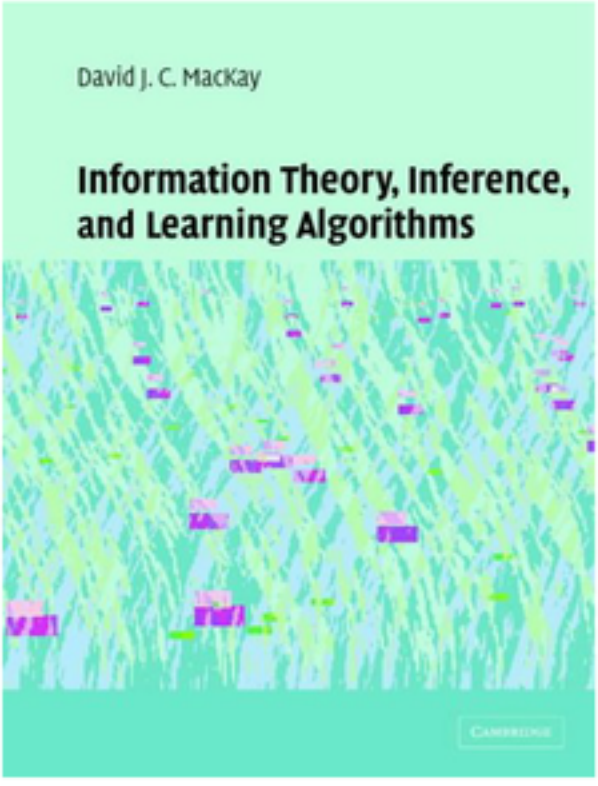


Information theory, pattern recognition, and neural networks

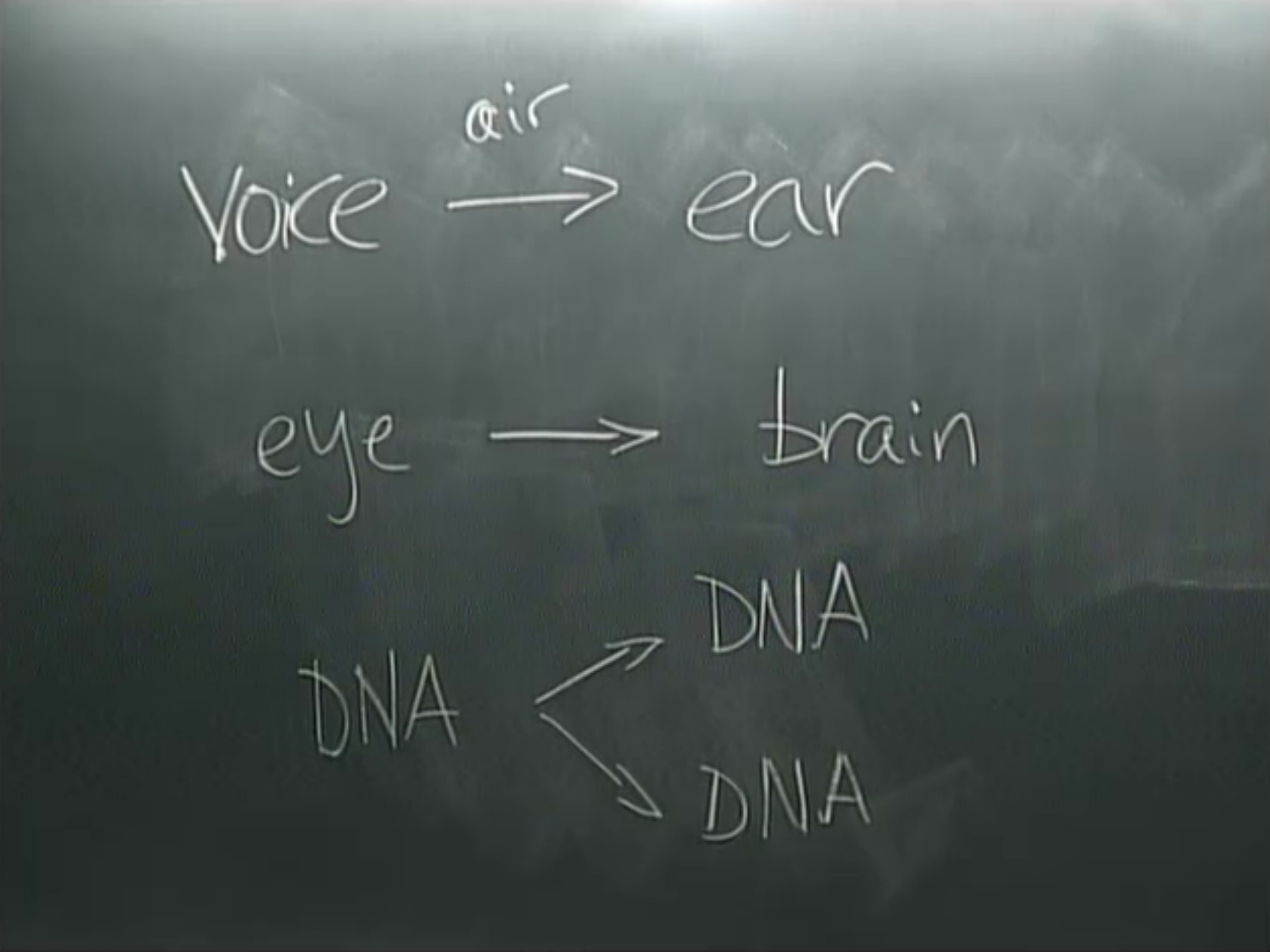
Lecture 1: Introduction to Information Theory

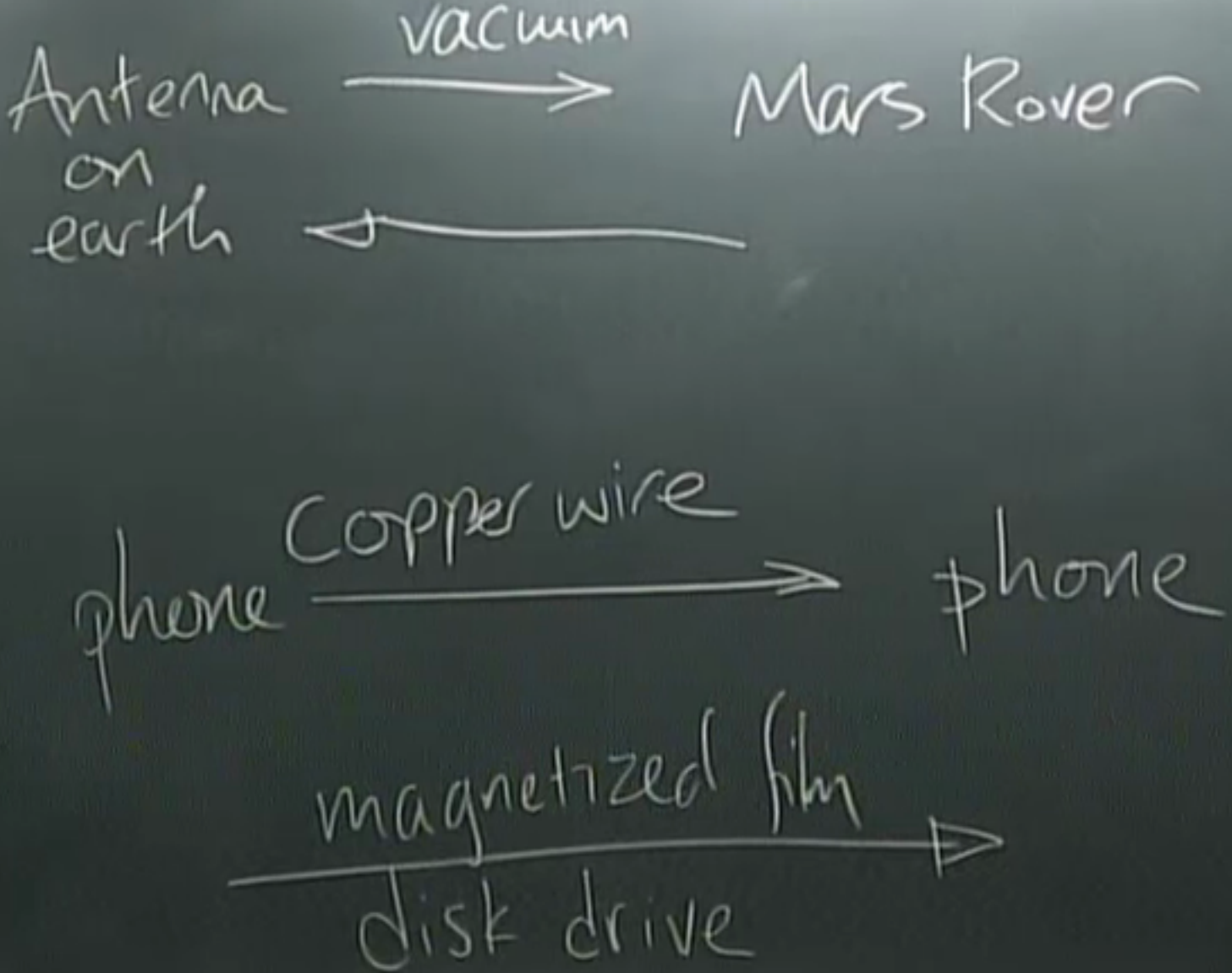
Lecture notes - Chapter 1



- Cambridge University Press
- 640 pages, ~ 35 pounds
- Also available free online

www.inference.phy.cam.ac.uk/mackay/itila/


air
Voice → ear
eye → brain
DNA
DNA
DNA

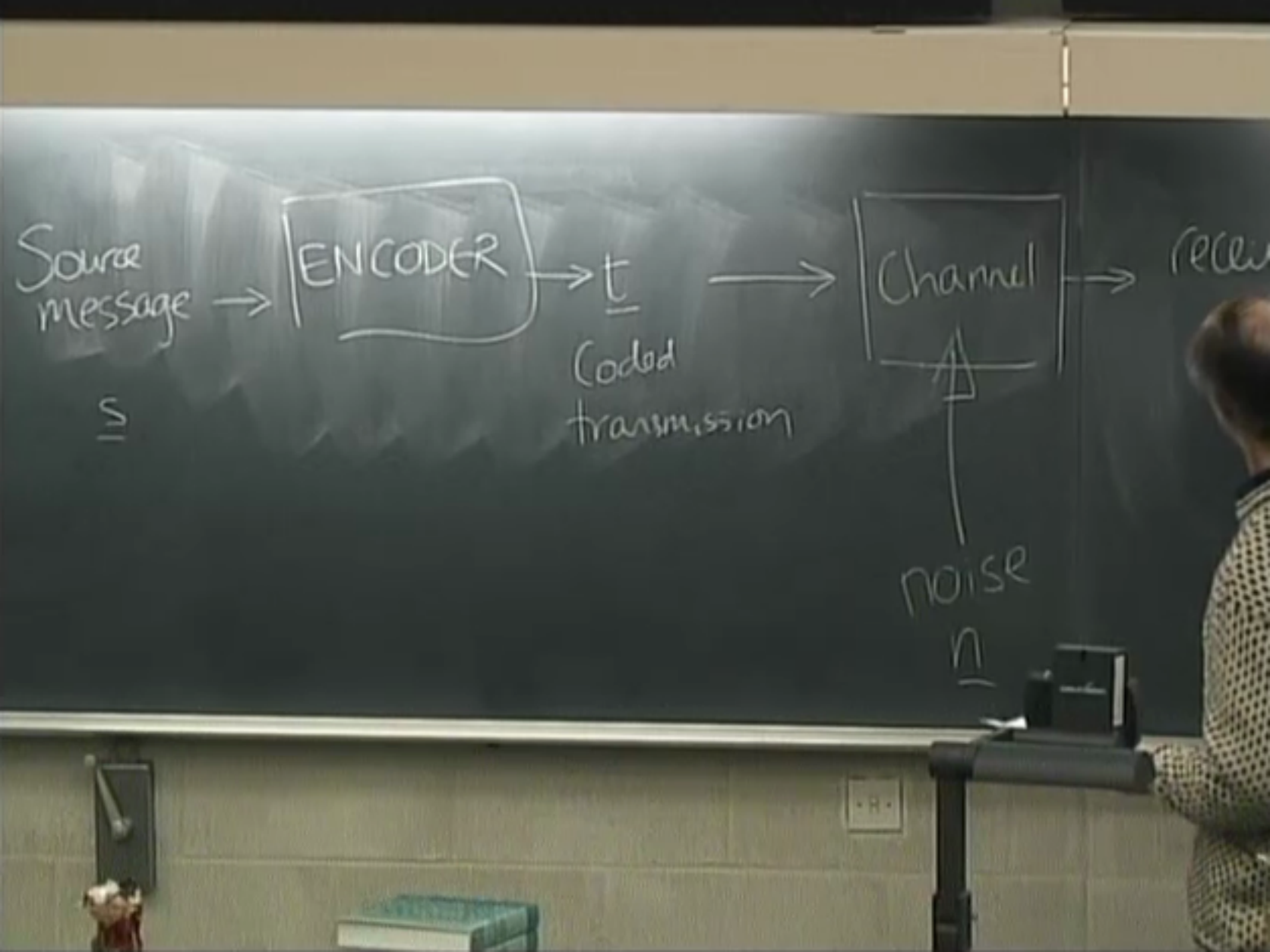

vacuum
Antenna on earth
Mars Rover
copper wire
phone
phone
magnetized film
disk drive

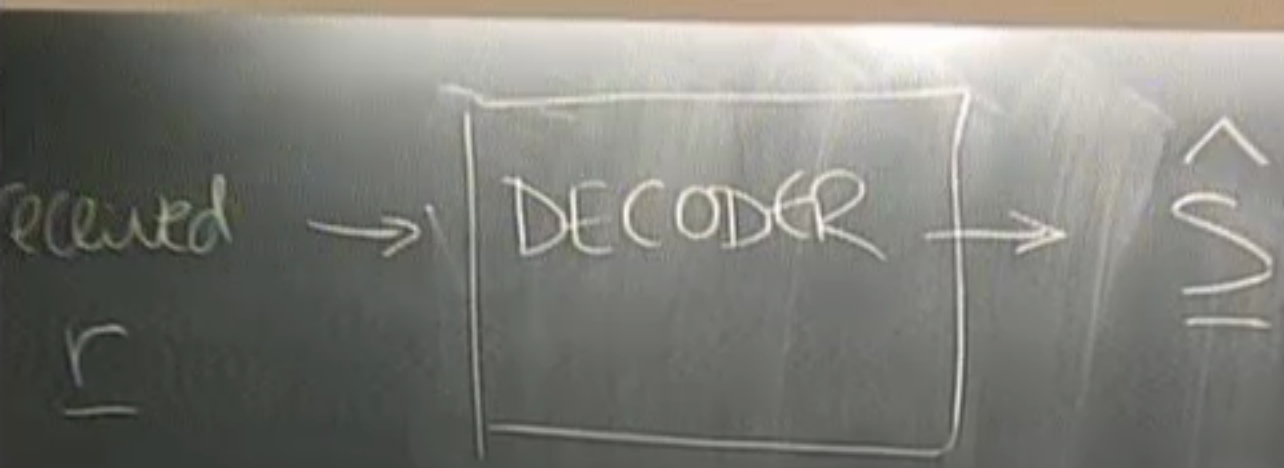
Received signal $\simeq$ Transmitted signal + noise

We would like

received message $=$ transmitted message

Solutions { Physical solutions
System solutions

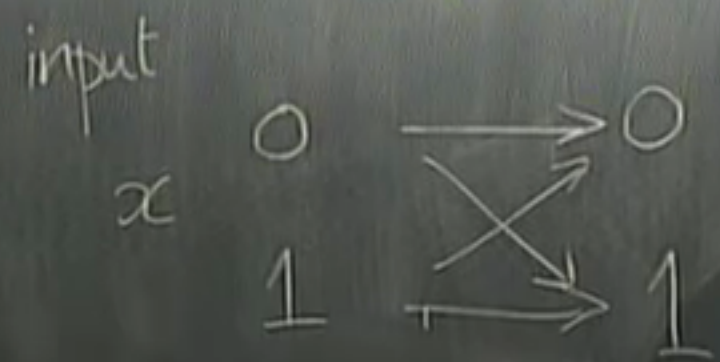

Source message
ENCODER
t
Coded transmission
Channel
noise
n
s



infers

$\underline{n}$ and $\underline{s}$

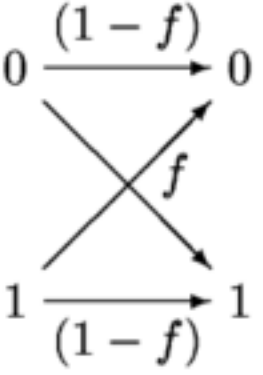
Binary symmetric channel



$P(y=0|x=0) = 1-f$

$P(y=1|x=0) = f$





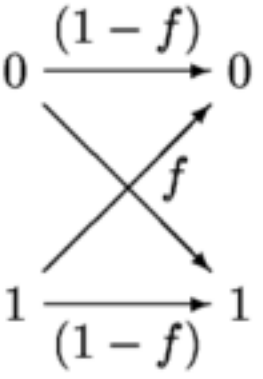


f = 0.1

$f = 0.1$

Q1: a file of $N = 10,000$ bits is stored, then read



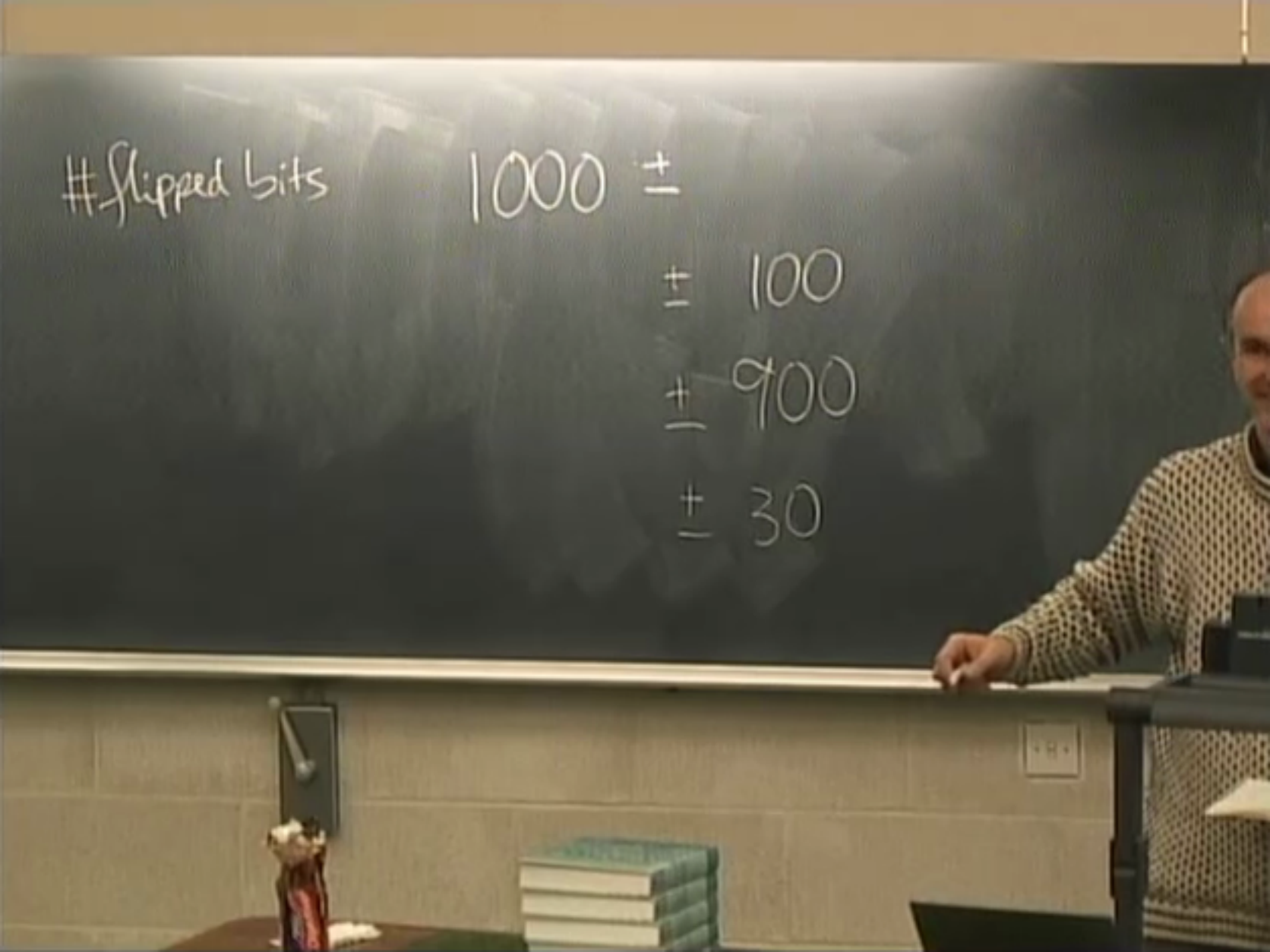


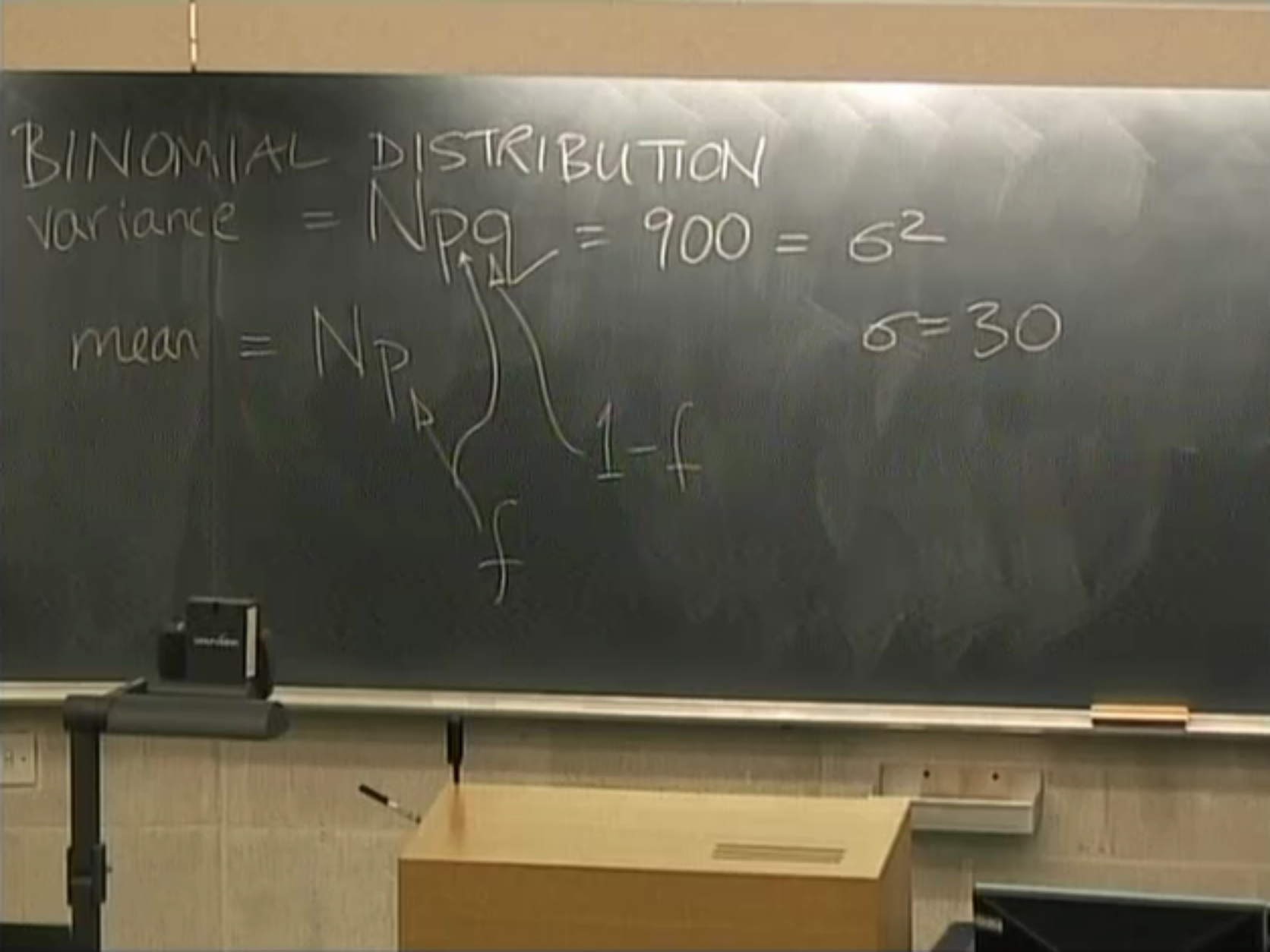


Q: A file of $N = 10\,000$ bits is stored on this disc drive (with $f = 0.1$), then read.

Roughly how many bits are flipped?

$$\boxed{} \pm \boxed{}$$


#flipped bits
1000 ±
± 100
± 900
± 30


BINOMIAL DISTRIBUTION
variance = NPq = 900 = σ²
σ = 30
mean = NP
1-f
f

Roughly how many bits are flipped?

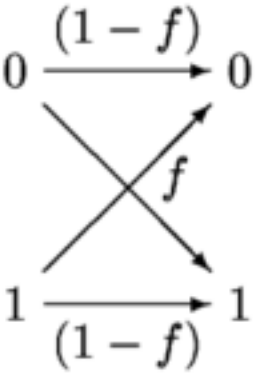
1000 ± 30

for a saleable

1 Gigabyte disk drive

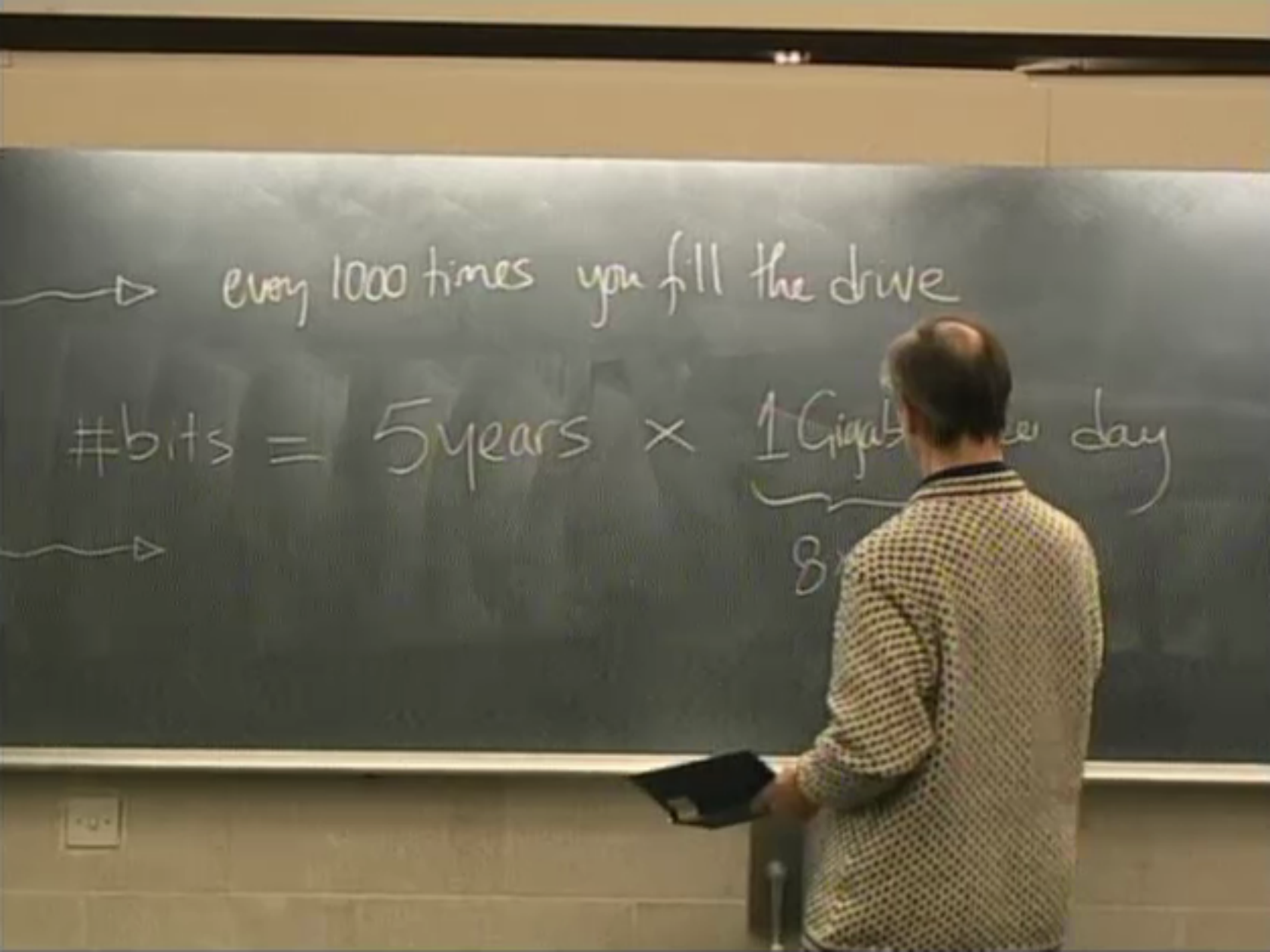
how small does f need to be?







Q: To make a successful business selling 1 Gigabyte disc drives, how small does the flip probability f need to be?


every 1000 times you fill the drive
#bits = 5years × 1
day

1000 times you fill the drive

5 years × 1 Gigabyte pe

$= 5 \times 365 \times 8 \times 10^9$ bits

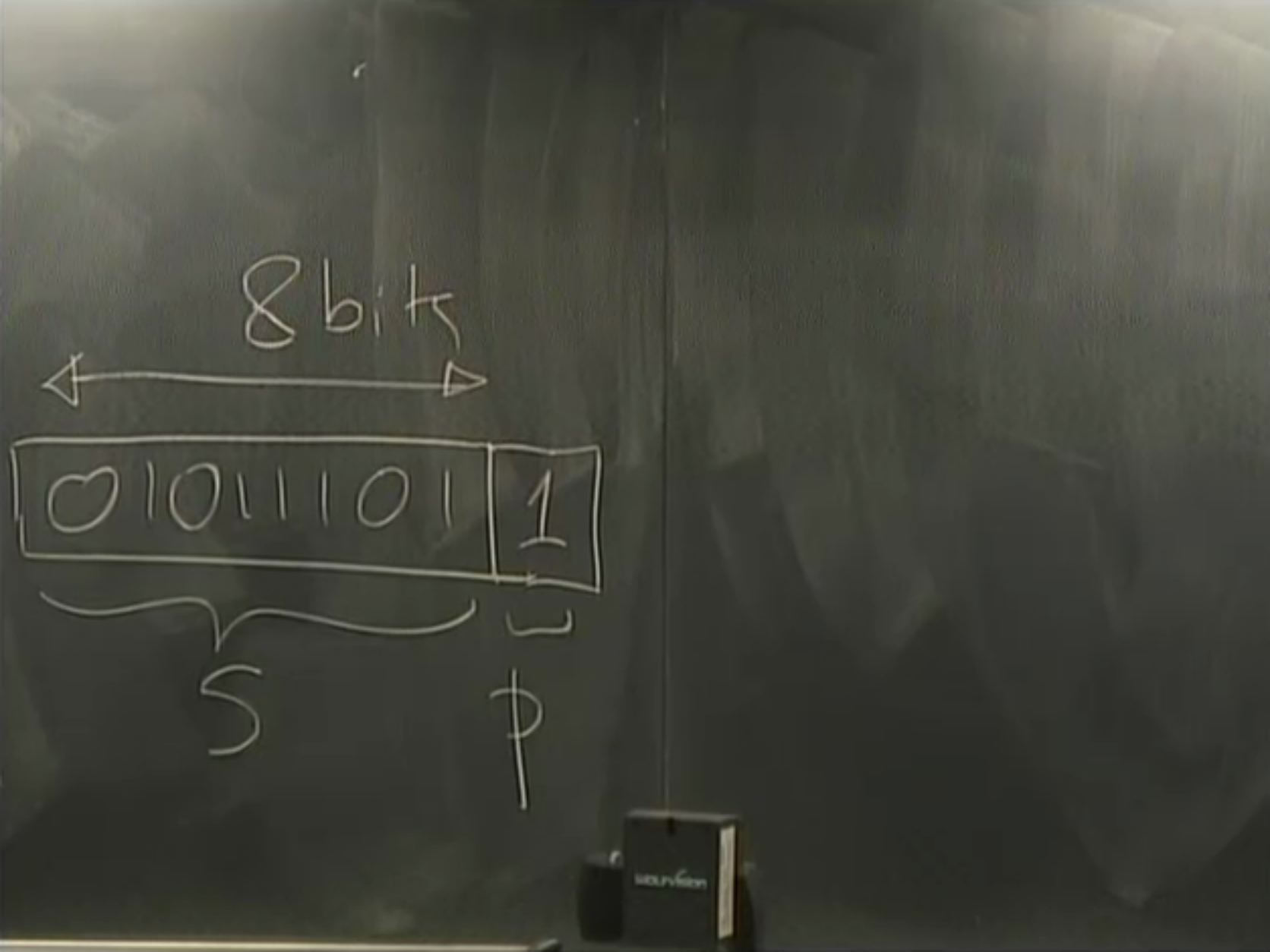
$\approx 10^{13}$

1% chance of disappointment

$f \simeq 10^{-15}$

1000 happy customers $\Rightarrow f \simeq 10^{-18}$

Aiming for 10^{-15} or better

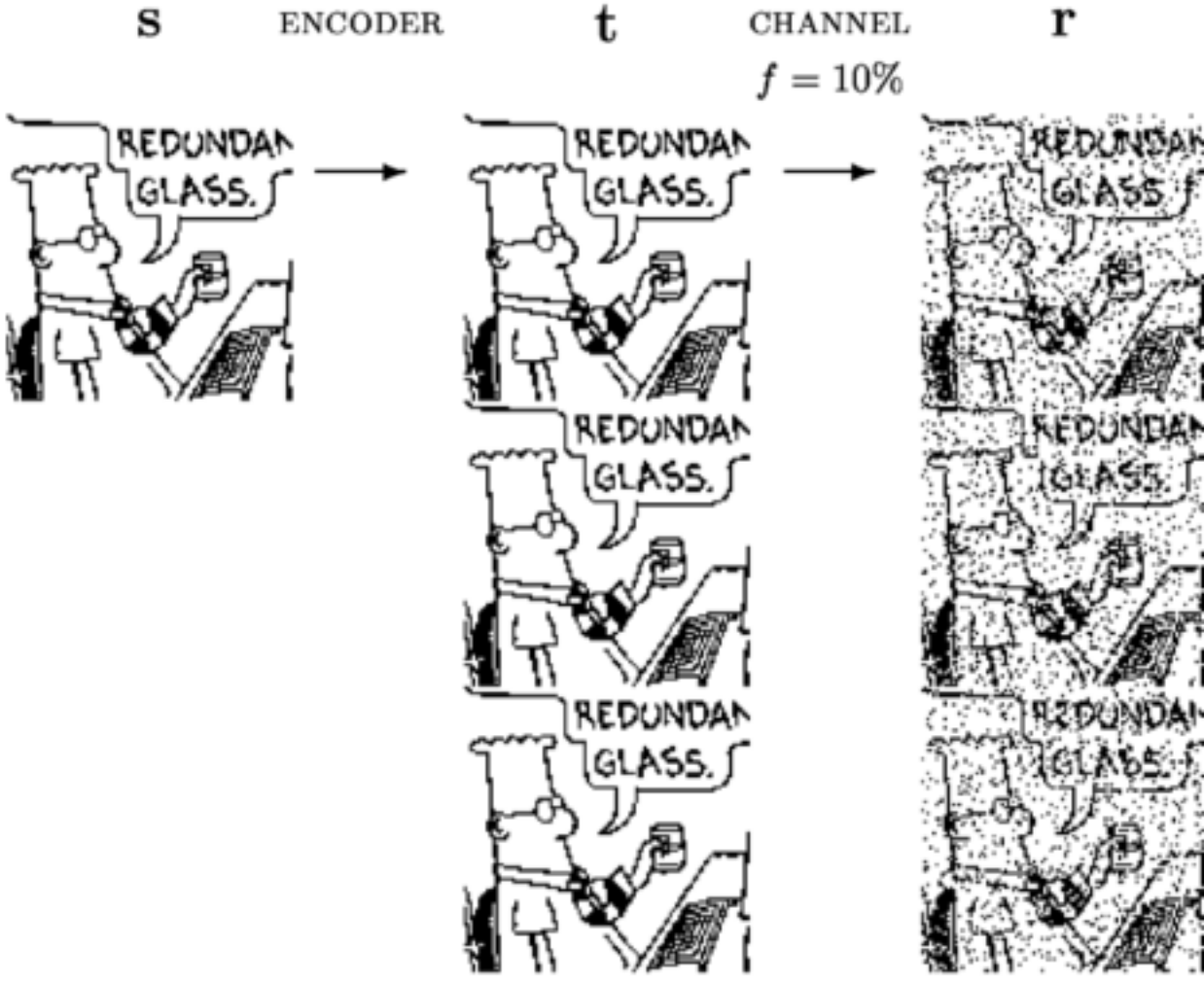

8bits
01011101
1
S
P

repetition code

R_3

s	t
0	$\rightarrow$ 000
1	$\rightarrow$ 111

Repetition code 'R3'

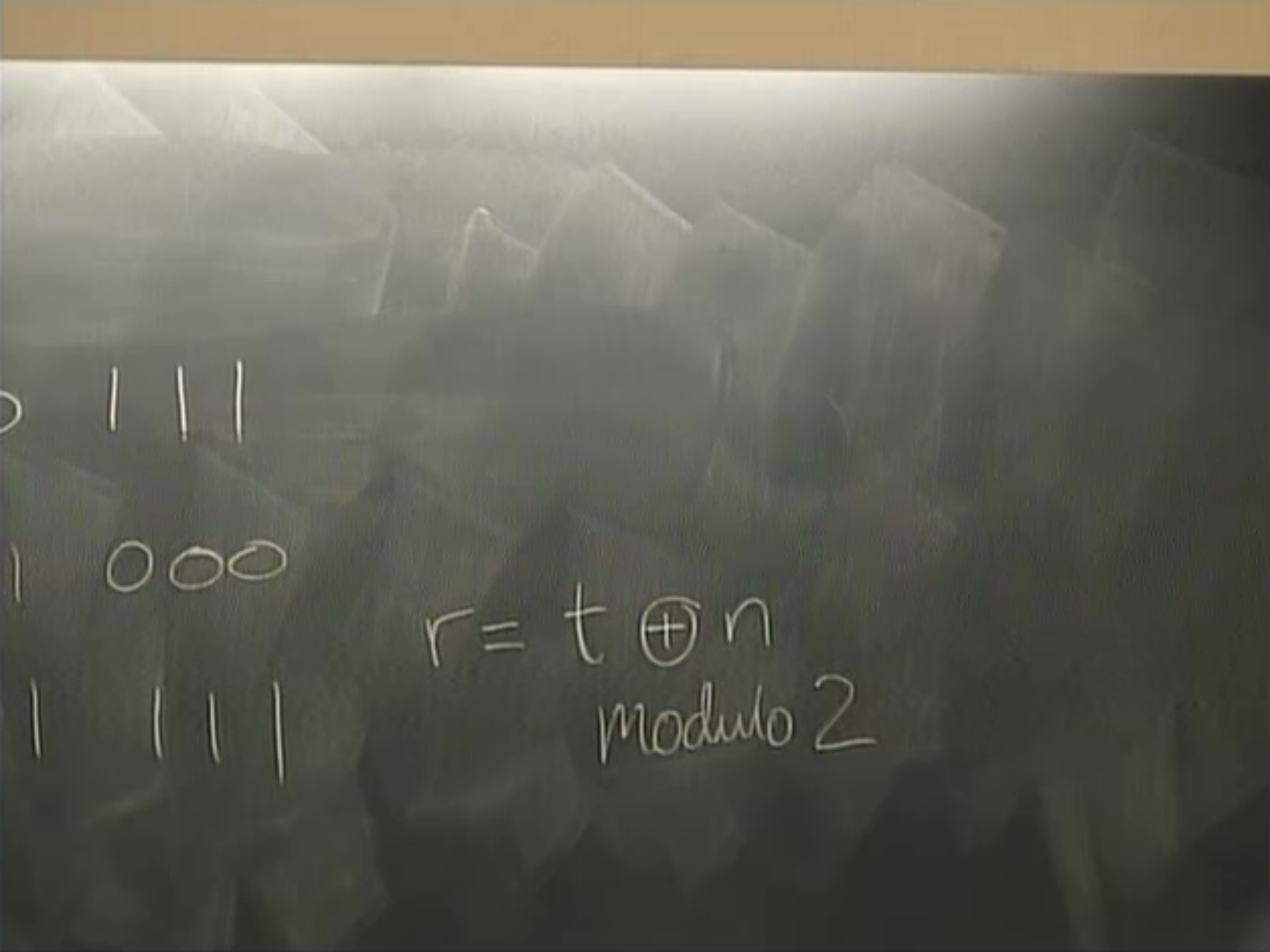


$\underline{s} = 01101$

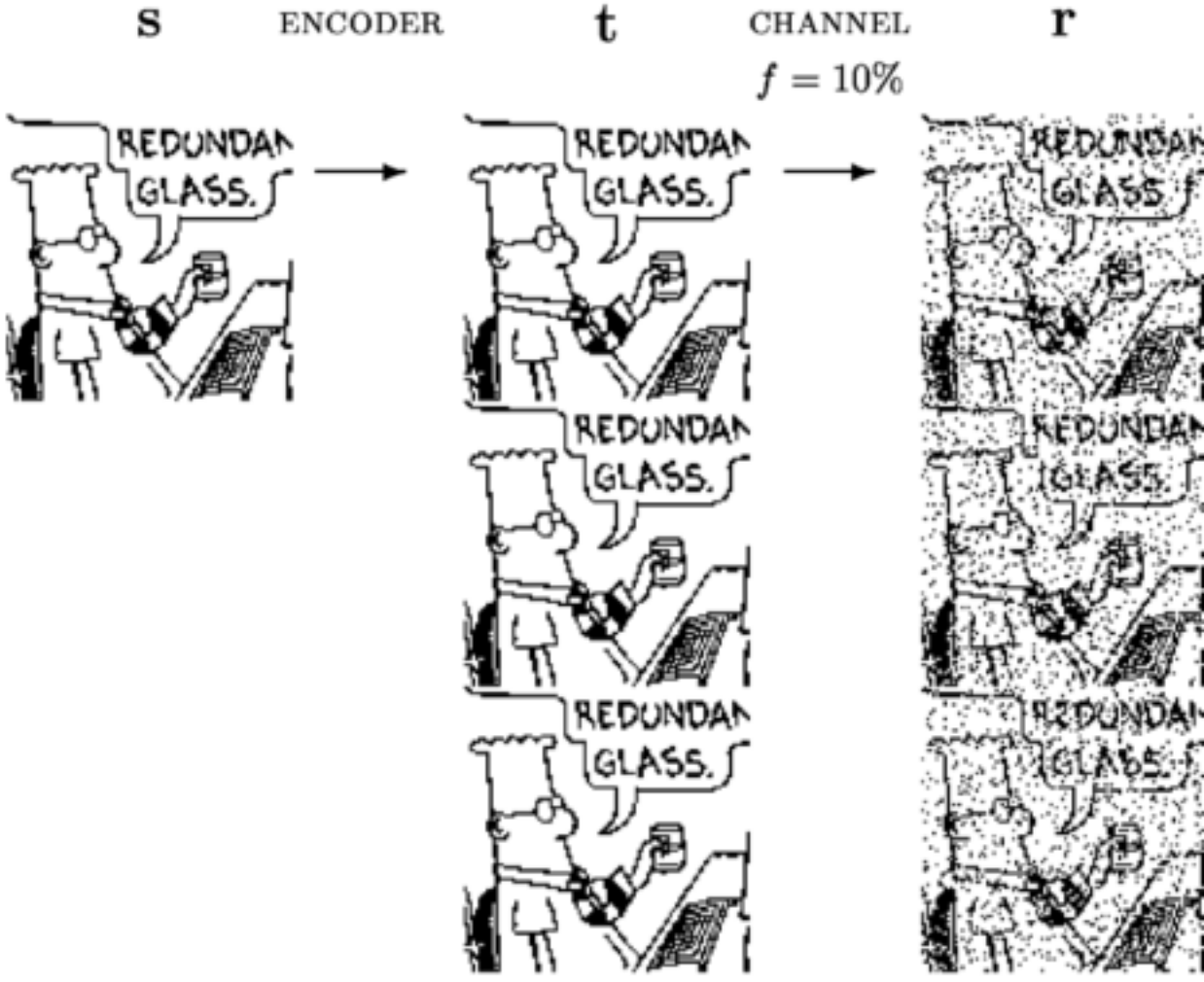
$t = 000\ 111\ 111\ 000\ 111$

$n = 000\ 100\ 000\ 101\ 000$

$r = 000\ 011\ 111\ 101\ 111$


1 1 1
1 0 0 0
1 1 1 1
r = t ⊕ n
modulo 2

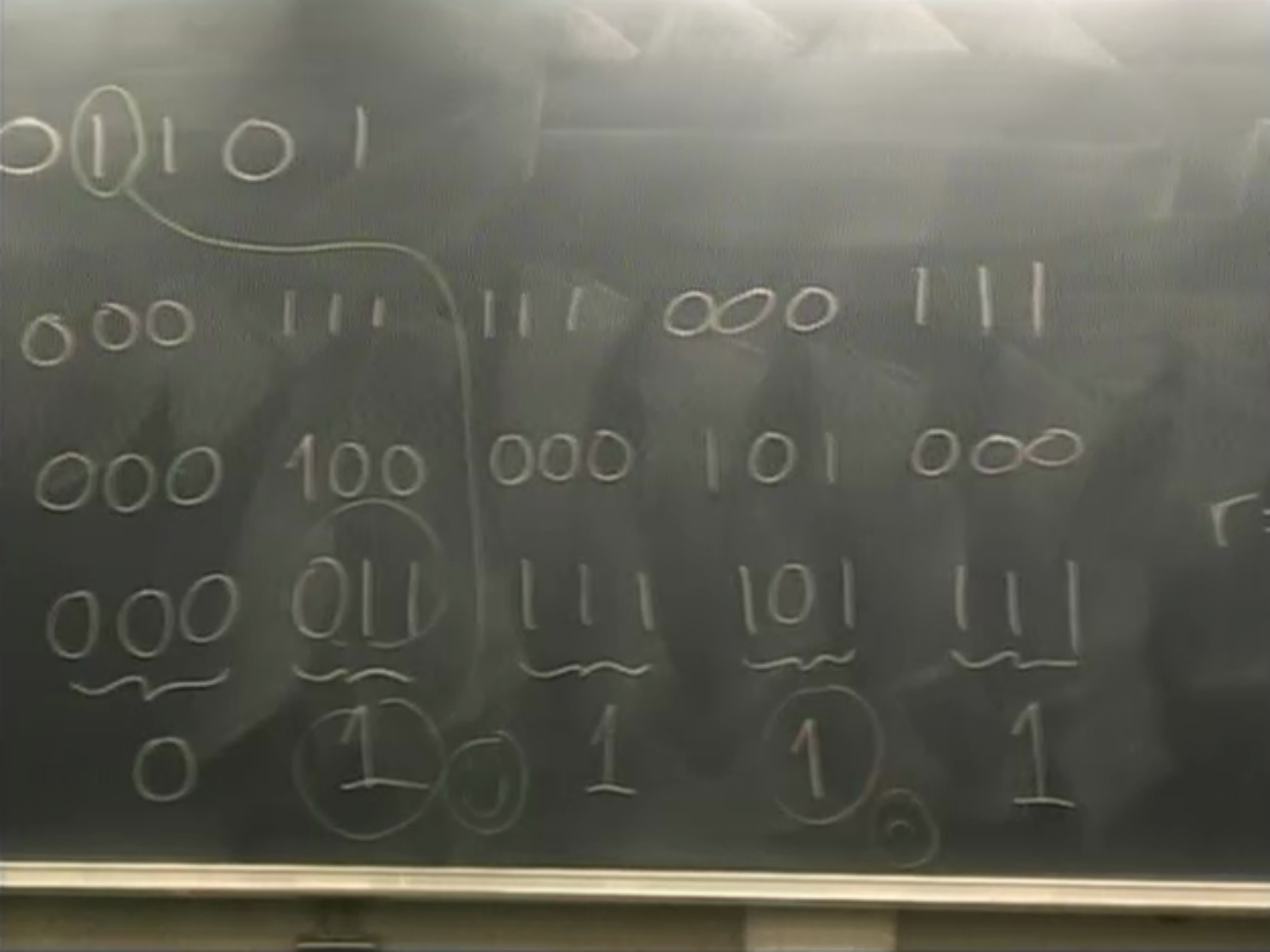
Repetition code 'R3'



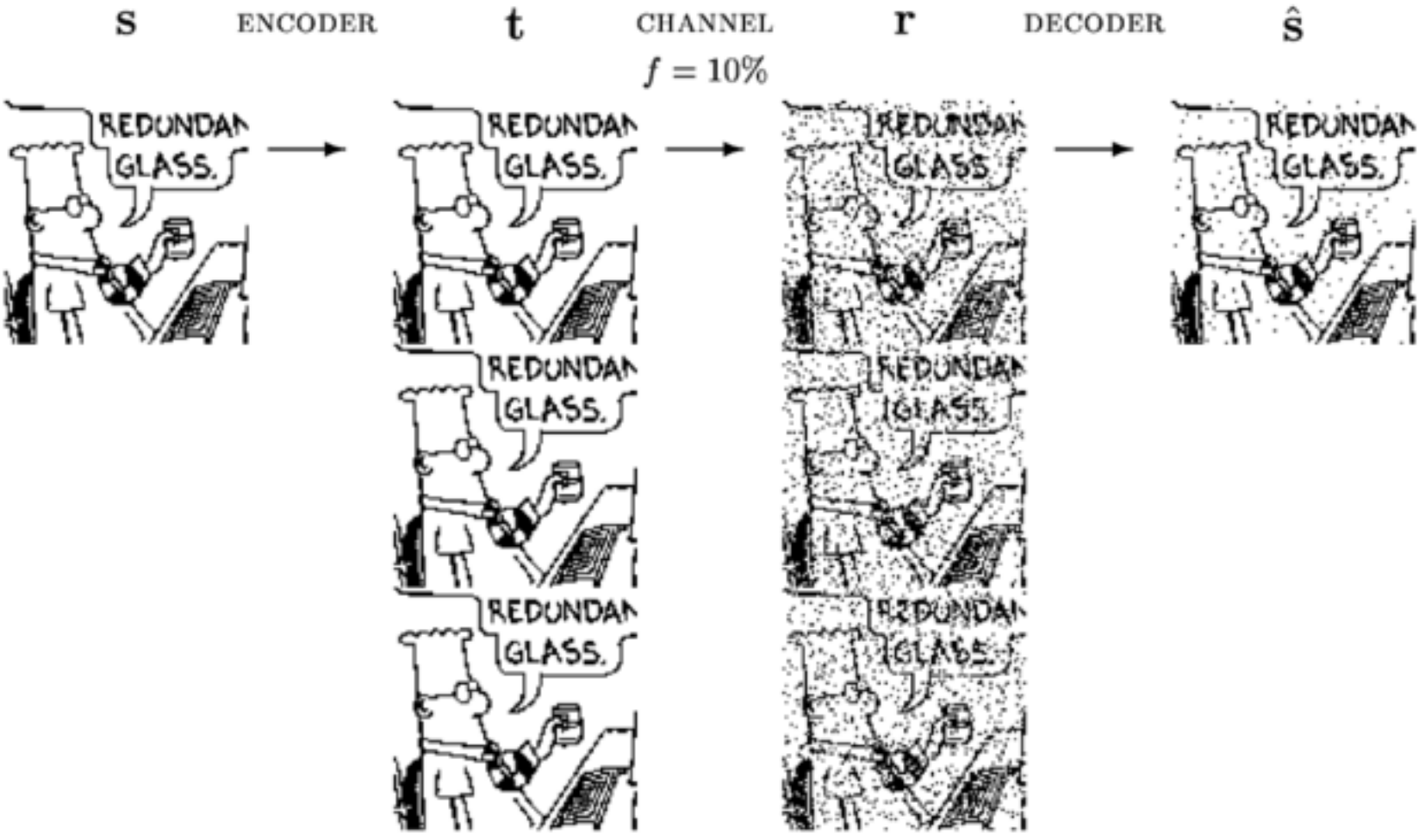
Decoder

r	$\rightarrow$	$\hat{s}$
000	$\rightarrow$	0
001	$\rightarrow$	0
⋮		
110	$\rightarrow$	1
111	$\rightarrow$	1

Best of
Best o
Major


0 1 1 0 1
0 0 0 1 1 1 1 1 1 0 0 0 1 1 1
0 0 0 1 0 0 0 0 0 1 0 0 0
0 0 0 0 1 1 1 1 1 1 0 1 1 1 1
0 1 0 1 1 1 0 1

Repetition code 'R3'



ference

Inverse probability

Product rule $P(s,r) = P(s)P(r|s)$

$= P(r)P(s|r)$

Sum rule $P(r) = \sum_s P(s,r) = P(s=$

Likelihood of s

Prior probability of s

$$\frac{P(r|s)\,P(s)}{P(r)}$$

sum rule

$$P(r) = P(r|s=0)P(s=0) + P(r|s=1)P($$

$r = 011$

$P(\underline{r} \mid s=0) = (1-f) \times f \times f$

$P(\underline{r} \mid s=1) = f \times (1-f) \times (1-f)$

$(1-f) \times f \times f = (1-f) f^2$

$f \times (1-f) \times (1-f) = f (1-f)$

1 ← # of bits that would have been flipped

$(1-f)\,f^2$

$= f\,(1-f)^2$

① = # of bits that would have been flipped

agreements between $\underline{t}(s)$ & $\underline{r}$

$P(s=0) = 1/2$

& $P(s=1) = 1/2$

then $P(s=1 \mid r=011) = \dfrac{(1-f)^2 f \cdot \frac{1}{2}}{(1-f)f^2\frac{1}{2} + f(1-f)^2\frac{1}{2}}$!

$$1) = \frac{(1-f)^2 f \cdot \frac{1}{2}}{(1-f) f^2 \frac{1}{2} + f (1-f)^2 \frac{1}{2}} = (1-f)$$

$\underline{r} \longrightarrow \hat{s}$

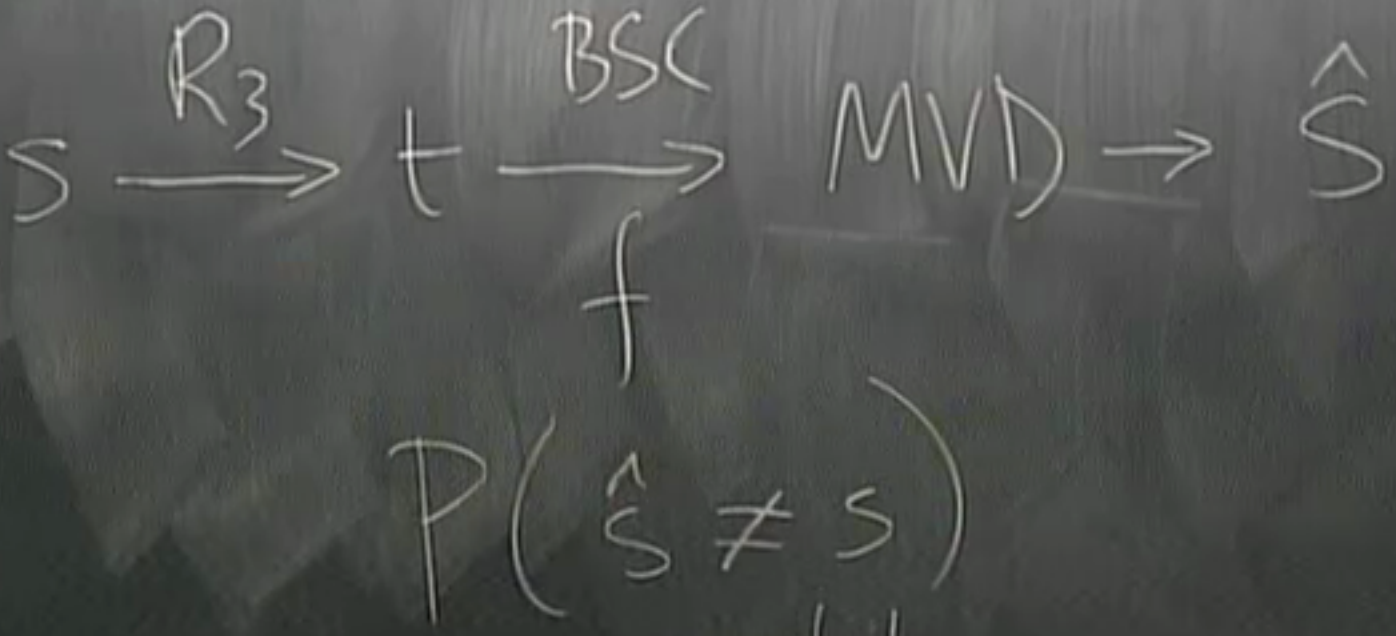
$011 \longrightarrow P(s=1 \mid r=011)$

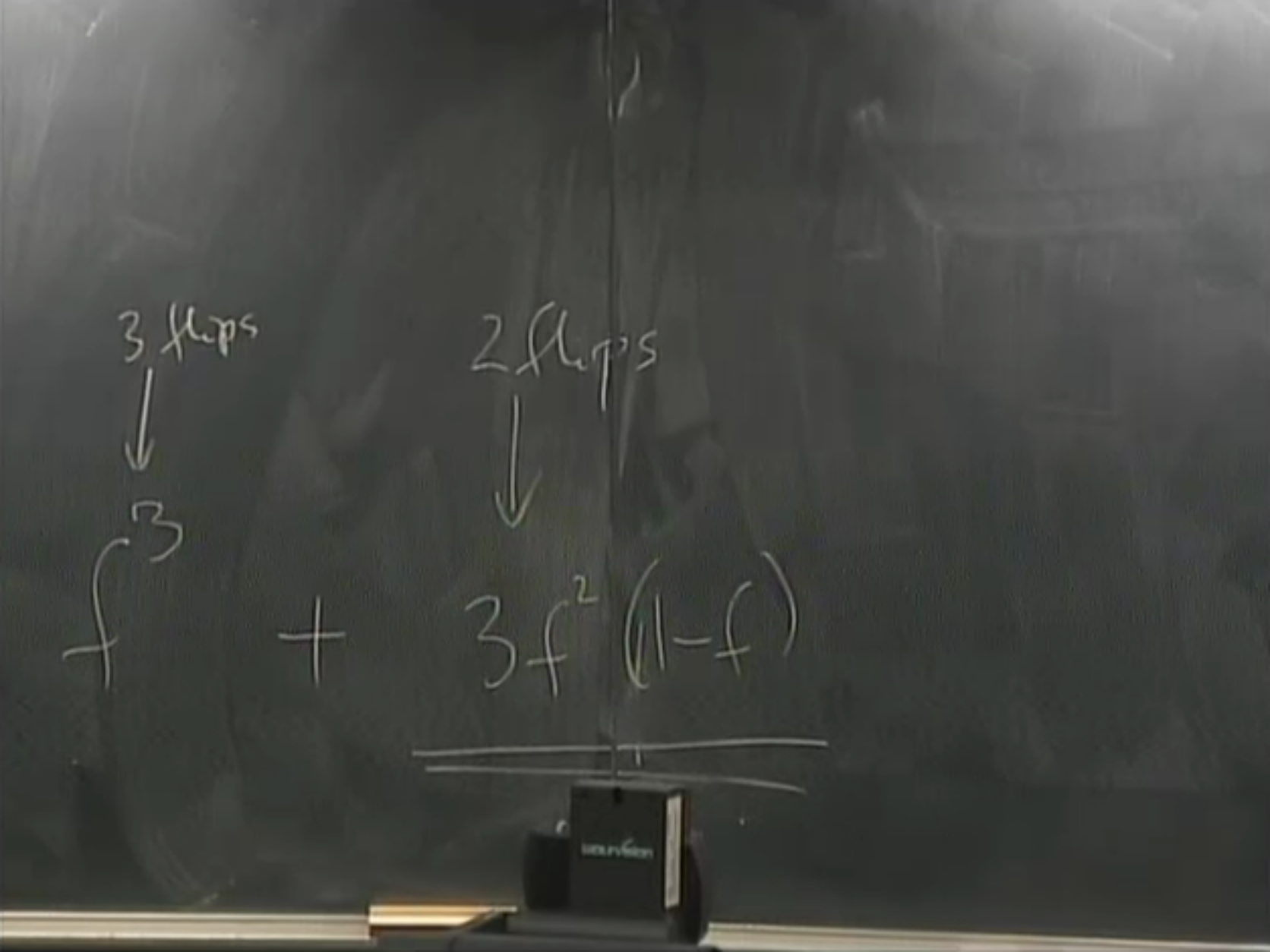
$= 1-f$

$P(s=1 \mid \underline{r}) > P(s=0 \mid r)$

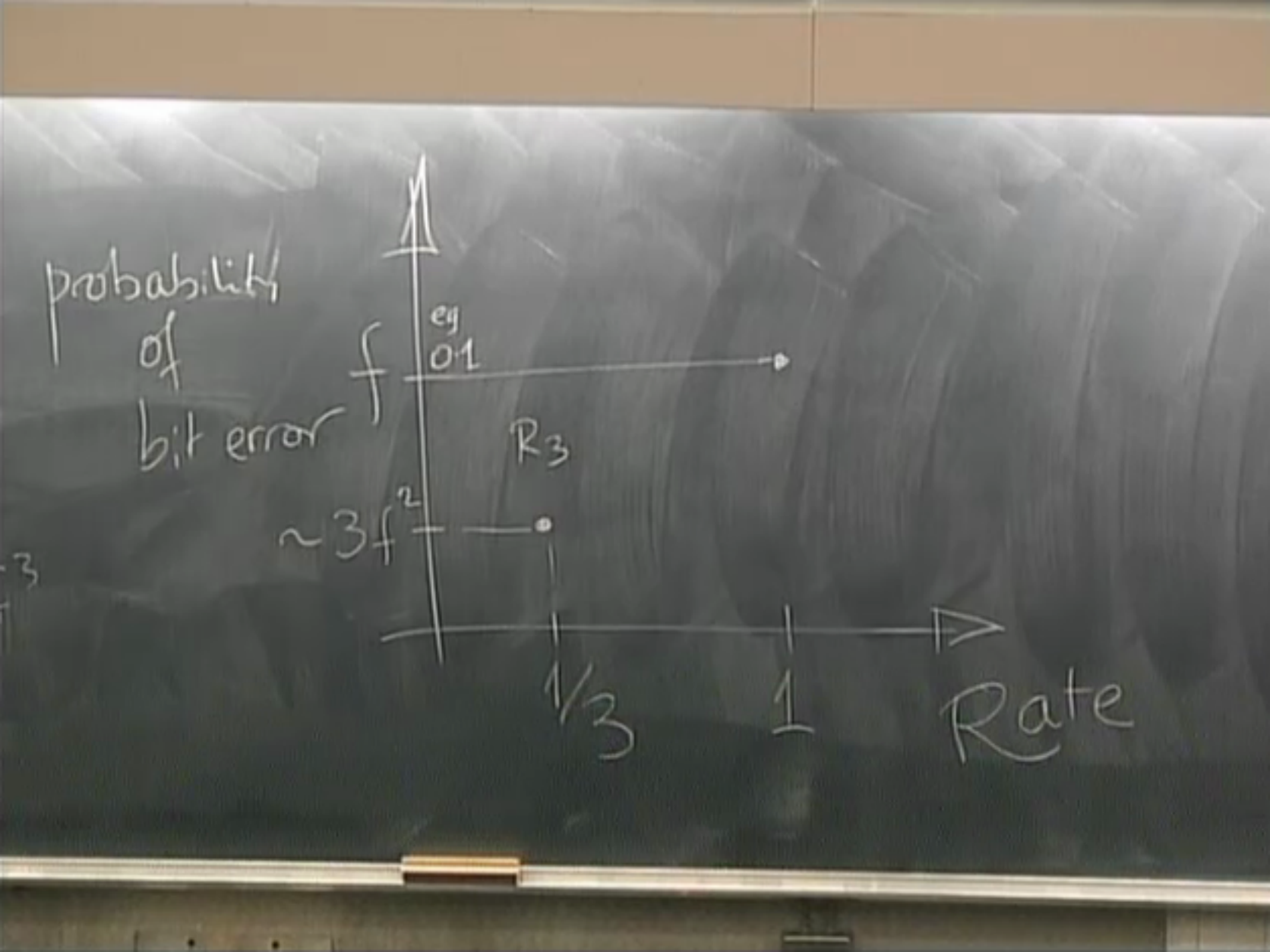
So $\hat{s}=1$ is best guess

Forward probability




3 flops
2 flop's
$f^3 + 3f^2(1-f)$

$$3f^2 - 2f^3$$


probability
of
bit error
f
eg
0.1
R3
~3f^2
1/3
1
Rate

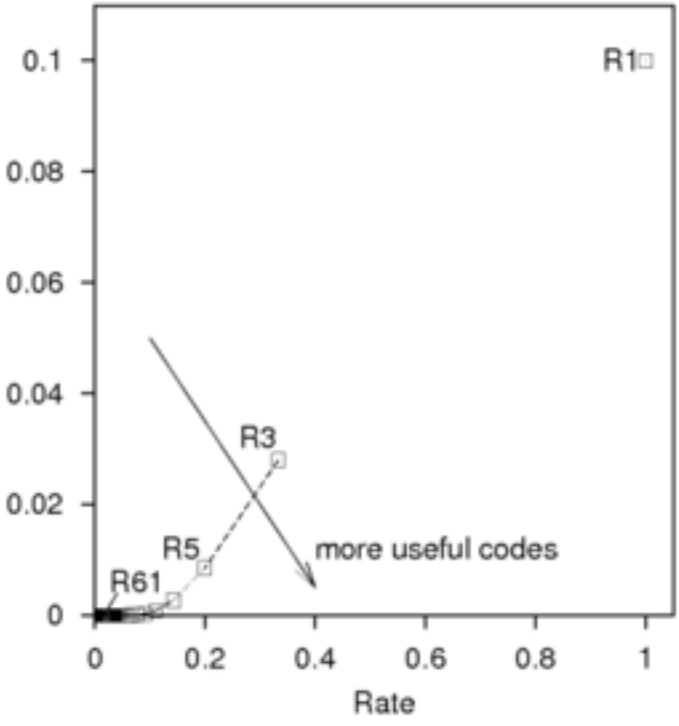
Hmk R_N

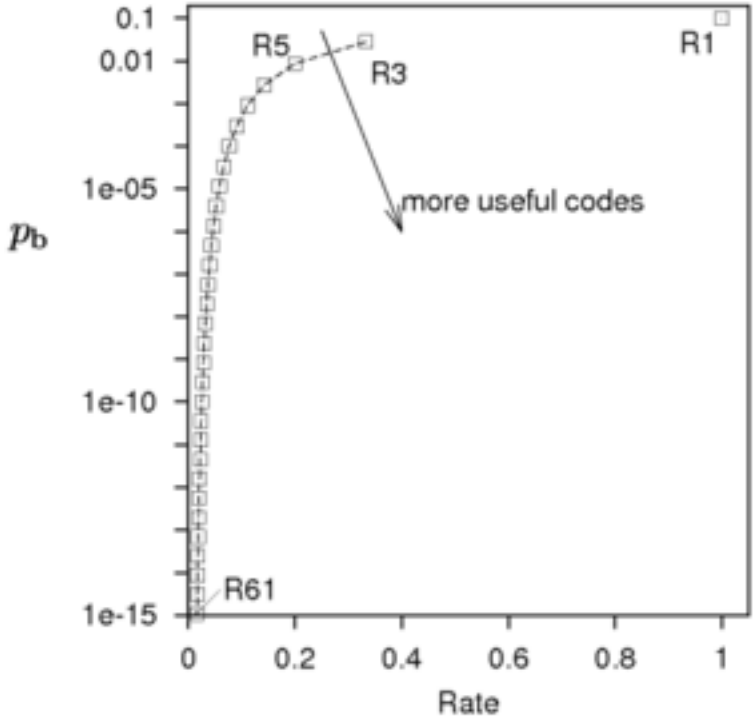
what N is needed to deliver $P_b \simeq 10^{-15}$

N = 61

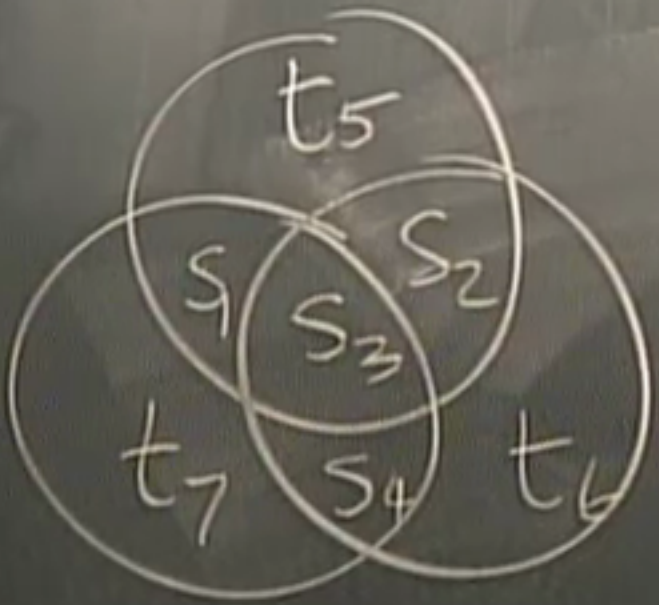
is
to deliver
0-15 ?

Performance of repetition codes

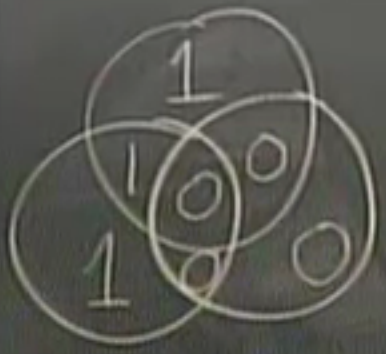




7,4 Hamming



Hamming Code

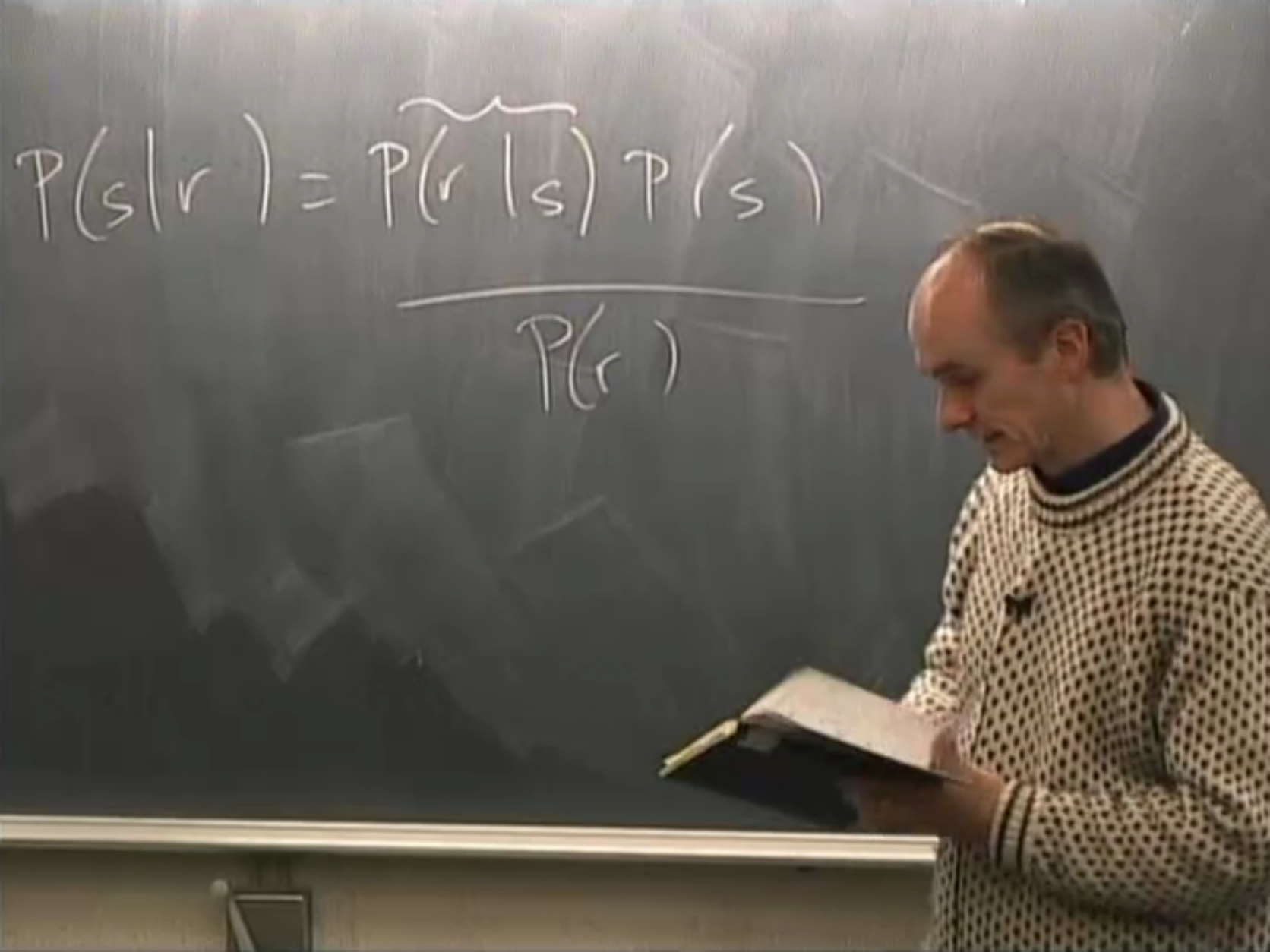


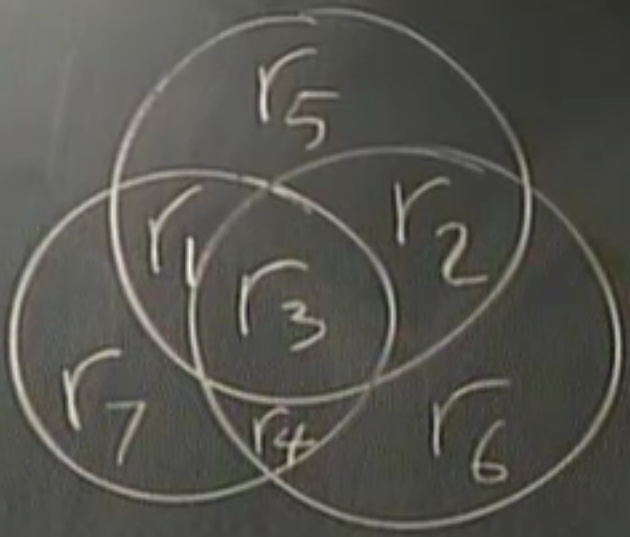
4 → 7

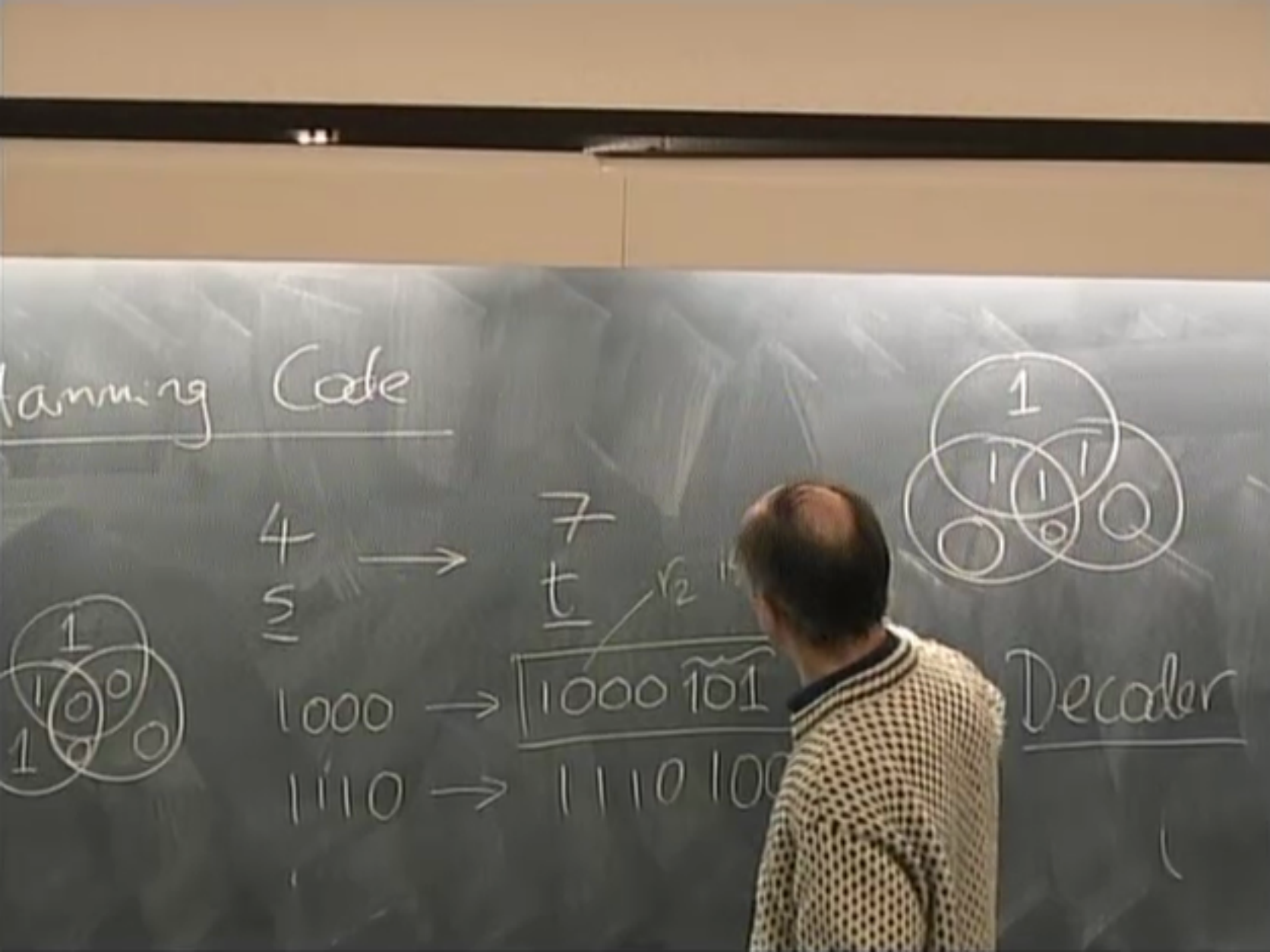
s → t

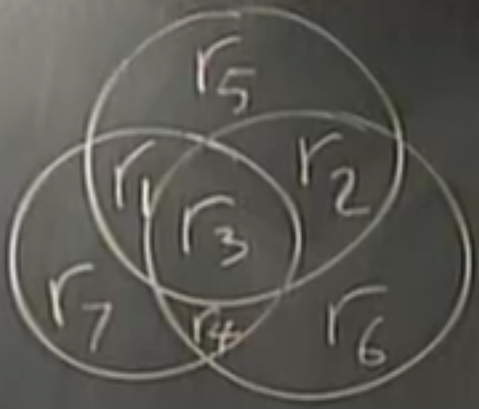
1000 → 1000 101

1110 → 1110 100


P(s|r) = P(r|s) P(s)
P(r)


r5
r1
r2
r3
r7
r4
r6


Hamming Code
4 → 7
s t
1000 → 1000 101
1110 → 1110 100
Decoder



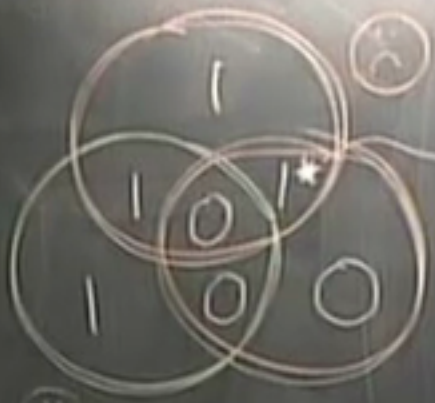
r = 1100101

=

pred

$\rightarrow \hat{t} = 1000101$

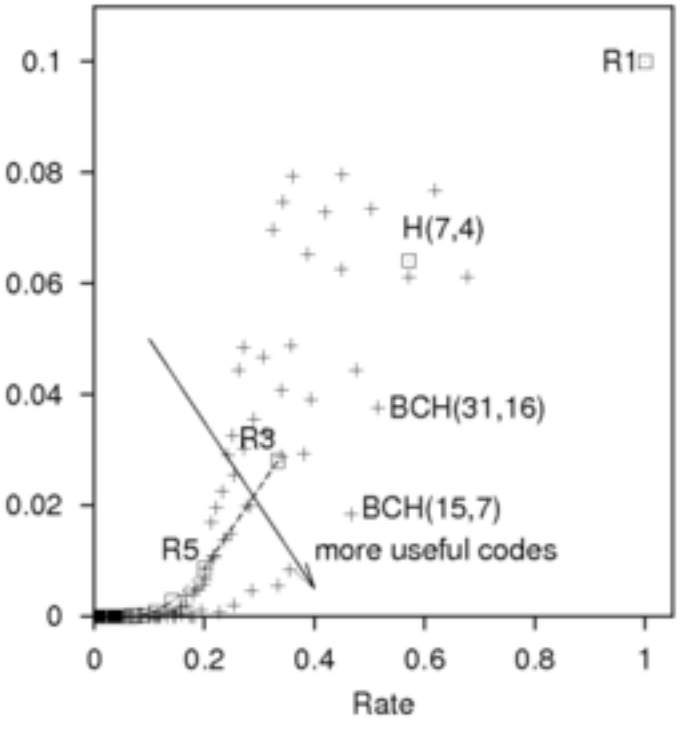
$\hat{s} = 1000$

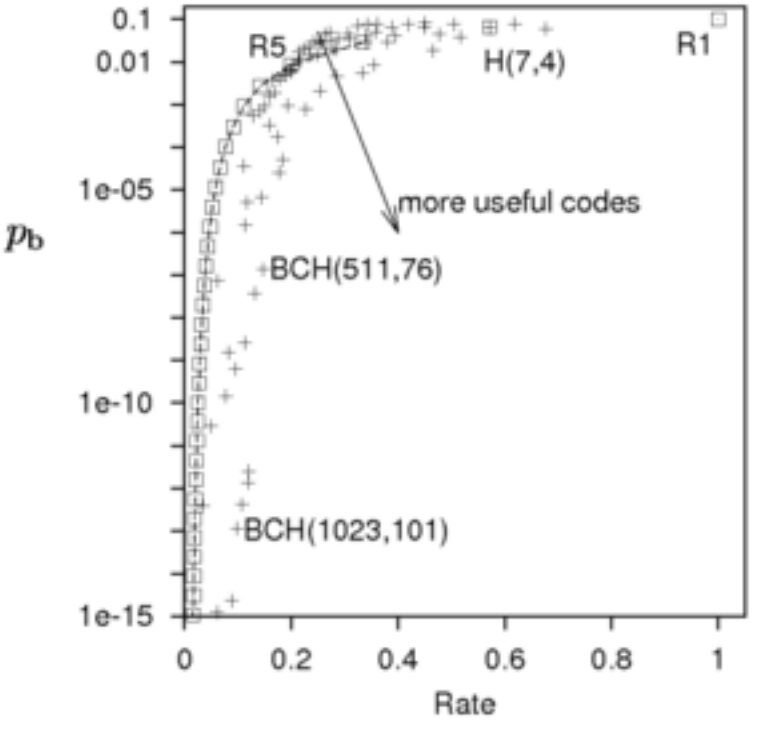


guess
this
was
flipped

syndrome → $\hat{t} = 1000$

$\hat{s} = 1000$

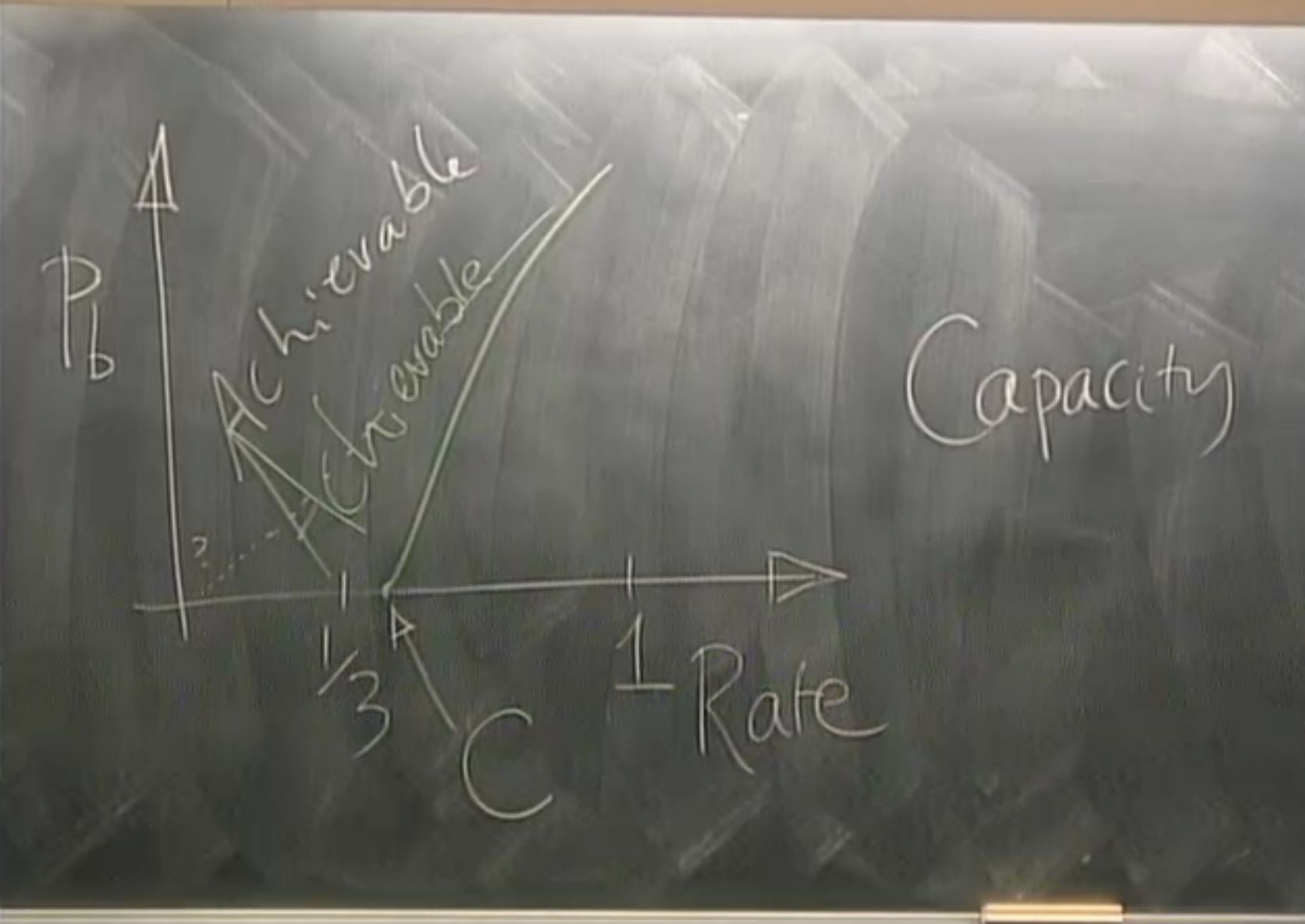


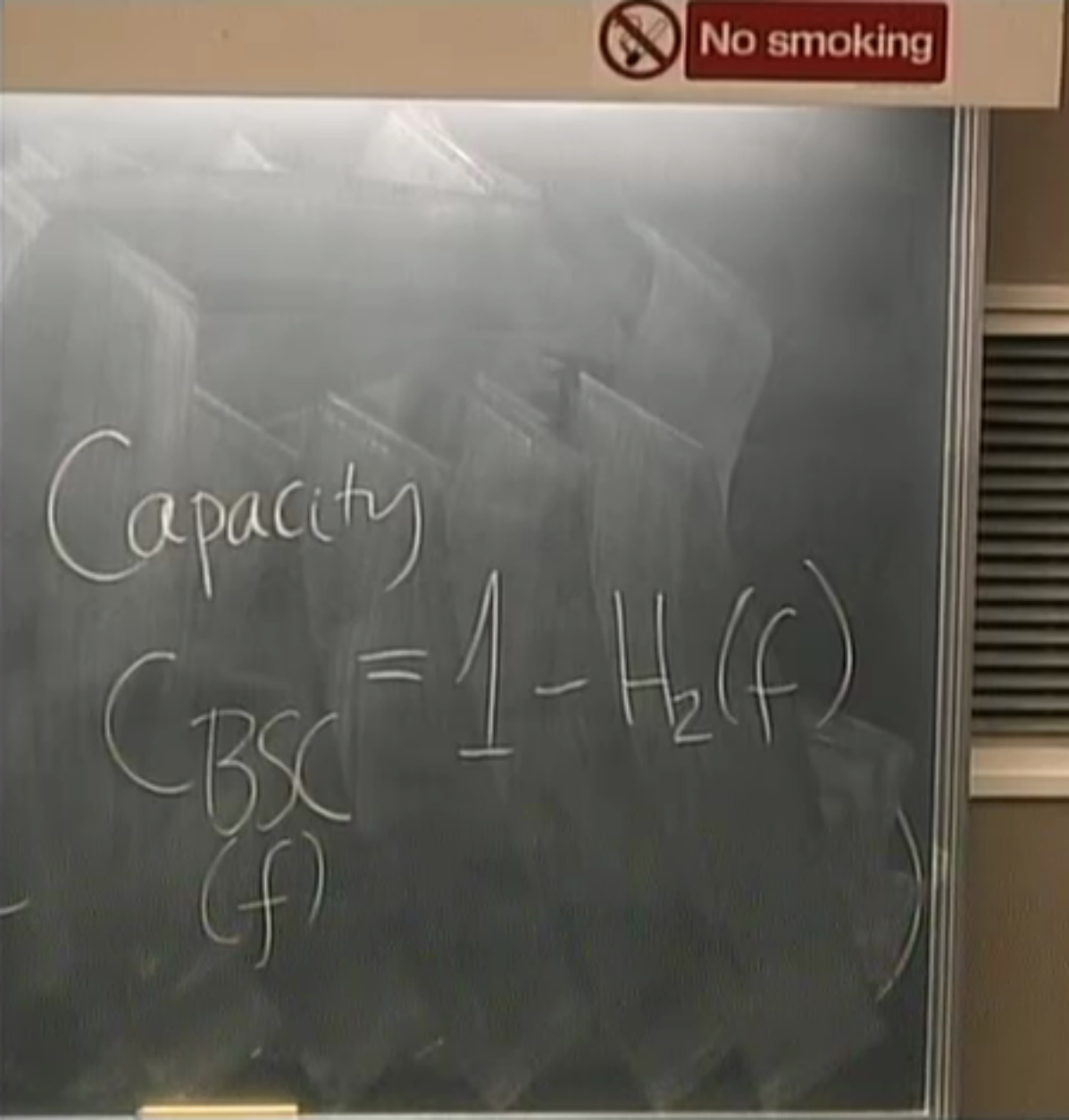


What's achievable?

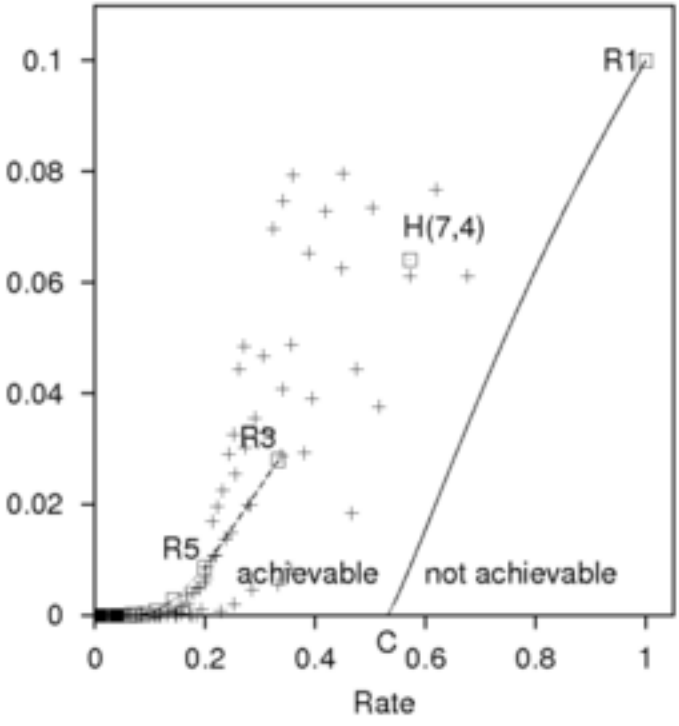
$$P_{\text{Block error}} \simeq 21 f^2 + \ldots$$

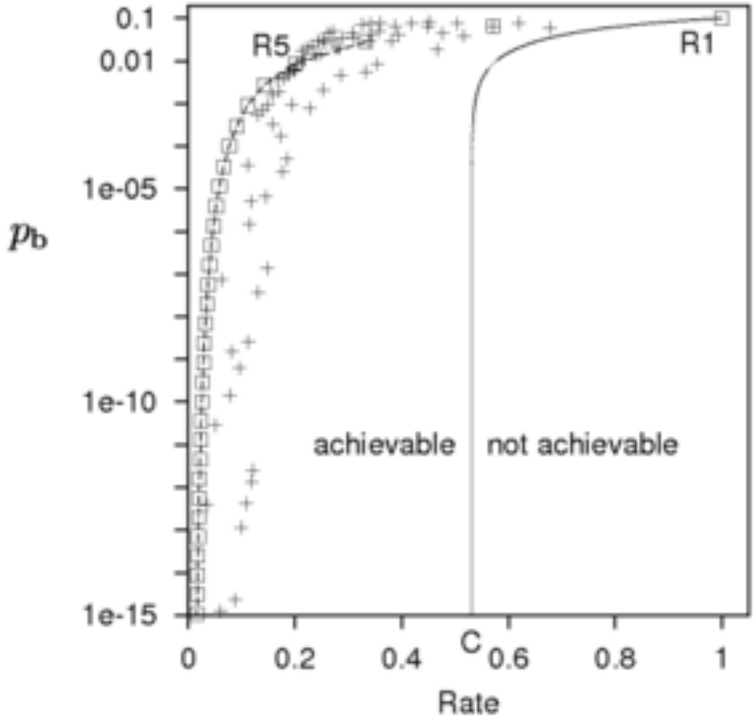
$$P_b \simeq 9 f^2 + \ldots$$


P_b
Achievable
Achievable
Capacity
1/3
C
1
Rate


No smoking
Capacity
$C_{BSC}(f) = 1 - H_2(f)$

Shannon's noisy-channel coding theorem

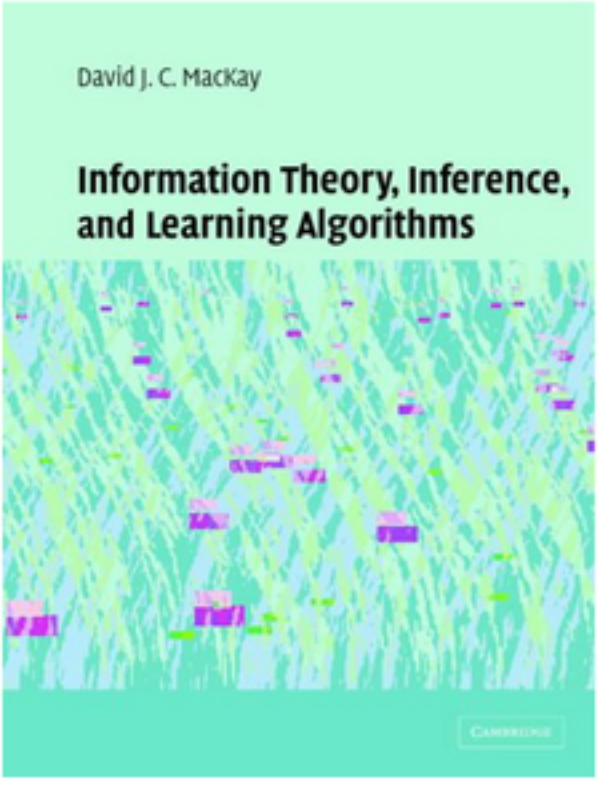




$$C_{\mathrm{BSC}}(f) = 1 - H_2(f)$$

$$H_2(f) = f \log_2 \frac{1}{f} + (1-f) \log_2 \frac{1}{1-f}$$

Lecture notes - Chapter 1

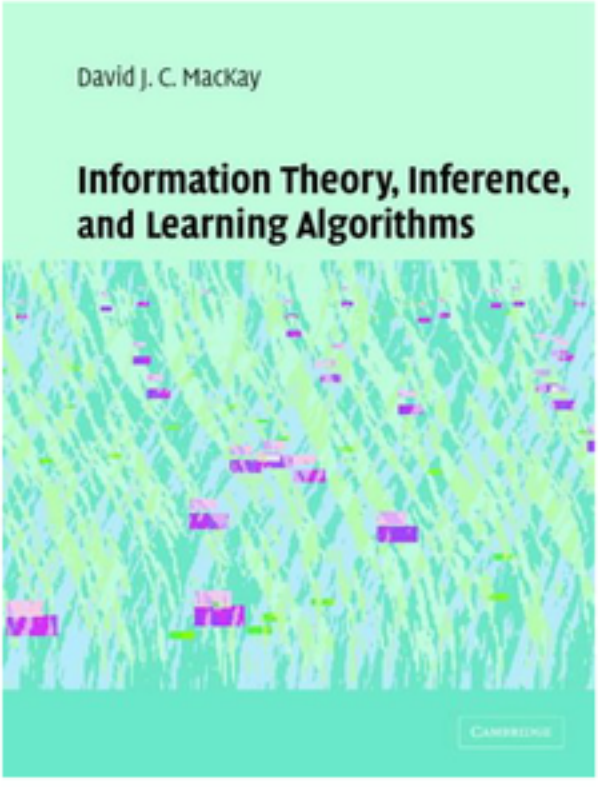


- Cambridge University Press
- 640 pages, ~ 35 pounds
- Also available free online

- Reading: Chapter 2

www.inference.phy.cam.ac.uk/itprnn/

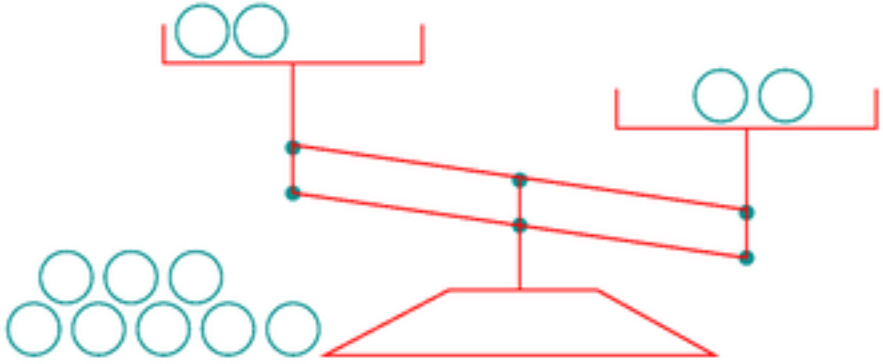
Course outline



- Information Theory
  - Source coding (Data compression)
  - Noisy-channel coding
- Data modelling, pattern recognition, learning & memory
  - Tools
    - Exact inference methods
    - Monte Carlo methods
    - Variational methods
  - Neural networks

www.inference.phy.cam.ac.uk/itprnn/

The weighing problem



You are given 12 balls, all equal in weight except for one that is either heavier or lighter.

Design a strategy to determine

which is the odd ball

and whether it is heavier or lighter,

in as few uses of the balance as possible.